

FAST FACTS

- The superheated water erupting from black smokers can have a temperature of 662°F (350°C), but it billows into water that is close to freezing.
- The smokelike effect is caused by chemicals that have turned into solid black particles. But other chemicals stay dissolved, allowing them to be absorbed by microbes and animals.
- Other vents in the ocean floor produce methane gas, and are home to similar forms of marine life. The seeping gas can freeze solid when it meets the cold water.

GIANT WORMS

The most spectacular animals living around black smokers and similar volcanic vents are giant tube worms, which can be up to 6 ft (1.8 m) long. The worms form dense colonies around the vents, where they can absorb the chemically rich water. They supply these chemicals to colonies of microbes living in their bodies, and the worms absorb some of the food made by the microbes. This is similar to the system used by the clams and mussels. It allows the worms to grow amazingly quickly, reaching full size in just a few months.

▼ RED PLUMES

Each worm lives in a thin-walled white tube. It has a plume of bright red feathery gills that it uses to absorb oxygen and vital chemicals.

